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##
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## mer. 1. févr. 13:21:30 2017
## Generated by MIG Version 3.92
##
#####
## File name:    example_top.ucf
##
## Details:  Constraints file
##          FPGA family:  spartan6
##          FPGA:        xc6slx100-fgg676
##          Speedgrade:   -3
##          Design Entry:  VHDL
##          Design:       with Test bench
##          DCM Used:     Enable
##          No.Of Memory Controllers: 1
##
#####

#####
# DDR2 requires the MCB to operate in Extended performance mode with higher Vccint
# specification to achieve maximum frequency. Therefore, the following CONFIG constraint
# follows the corresponding GUI option setting. However, DDR3 can operate at higher
# frequencies with any Vccint value by operating MCB in extended mode. Please do not
# remove/edit the below constraint to avoid false errors.
#####
CONFIG MCB_PERFORMANCE= EXTENDED;

#####
# Timing Ignore constraints for paths crossing the clock domain
#####
NET "memc?_wrapper_inst/memc?_mcb_raw_wrapper_inst/selfrefresh_mcb_mode" TIG;
NET "c?_pll_lock" TIG;
INST
"memc?_wrapper_inst/memc?_mcb_raw_wrapper_inst/gen_term_calib.mcb_soft_calibration_top_inst/mcb_soft_calibration_
inst/DONE_SOFTANDHARD_CAL*" TIG;

#Please uncomment the below TIG if used in a design which enables self-refresh mode
#NET
"memc?_wrapper_inst/memc?_mcb_raw_wrapper_inst/gen_term_calib.mcb_soft_calibration_top_inst/mcb_soft_calibration_
inst/SELFREFRESH_MCB_REQ" TIG;

#####
## Memory Controller 3
## Memory Device: DDR3_SDRAM->MT41J128M16XX-125
## Frequency: 400 MHz
## Time Period: 2500 ps
## Supported Part Numbers: MT41J128M16HA-125
#####
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#####
## Clock constraints
#####
NET "memc3_infrastructure_inst/sys_clk_ibufg" TNM_NET = "SYS_CLK3";
TIMESPEC "TS_SYS_CLK3" = PERIOD "SYS_CLK3" 2.5 ns HIGH 50 %;
#####

#####
## I/O TERMINATION
#####
NET "mcb3_dram_dq[*]"          IN_TERM = NONE;
NET "mcb3_dram_dqs"           IN_TERM = NONE;
NET "mcb3_dram_dqs_n"         IN_TERM = NONE;
NET "mcb3_dram_udqs"          IN_TERM = NONE;
NET "mcb3_dram_udqs_n"        IN_TERM = NONE;

#####
# I/O STANDARDS
#####

NET "mcb3_dram_dq[*]"          IOSTANDARD = SSTL15_II ;
NET "mcb3_dram_a[*]"          IOSTANDARD = SSTL15_II ;
NET "mcb3_dram_ba[*]"         IOSTANDARD = SSTL15_II ;
NET "mcb3_dram_dqs"           IOSTANDARD = DIFF_SSTL15_II ;
NET "mcb3_dram_udqs"          IOSTANDARD = DIFF_SSTL15_II ;
NET "mcb3_dram_dqs_n"         IOSTANDARD = DIFF_SSTL15_II ;
NET "mcb3_dram_udqs_n"        IOSTANDARD = DIFF_SSTL15_II ;
NET "mcb3_dram_ck"            IOSTANDARD = DIFF_SSTL15_II ;
NET "mcb3_dram_ck_n"          IOSTANDARD = DIFF_SSTL15_II ;
NET "mcb3_dram_cke"           IOSTANDARD = SSTL15_II ;
NET "mcb3_dram_ras_n"         IOSTANDARD = SSTL15_II ;
NET "mcb3_dram_cas_n"         IOSTANDARD = SSTL15_II ;
NET "mcb3_dram_we_n"          IOSTANDARD = SSTL15_II ;
NET "mcb3_dram_odt"           IOSTANDARD = SSTL15_II ;
NET "mcb3_dram_reset_n"       IOSTANDARD = LVCMOS15 ;
NET "mcb3_dram_dm"            IOSTANDARD = SSTL15_II ;
NET "mcb3_dram_udm"           IOSTANDARD = SSTL15_II ;
NET "mcb3_rzq"                IOSTANDARD = SSTL15_II ;
NET "mcb3_zio"                IOSTANDARD = SSTL15_II ;

#####
# MCB 3
# Pin Location Constraints for Clock, Masks, Address, and Controls
#####

NET "mcb3_dram_a[0]"          LOC = "N7" ;
NET "mcb3_dram_a[10]"         LOC = "M10" ;
NET "mcb3_dram_a[11]"         LOC = "L3" ;
NET "mcb3_dram_a[12]"         LOC = "M8" ;
NET "mcb3_dram_a[13]"         LOC = "M6" ;
NET "mcb3_dram_a[1]"          LOC = "N6" ;
NET "mcb3_dram_a[2]"          LOC = "R9" ;
NET "mcb3_dram_a[3]"          LOC = "P7" ;
NET "mcb3_dram_a[4]"          LOC = "N9" ;
NET "mcb3_dram_a[5]"          LOC = "R2" ;
NET "mcb3_dram_a[6]"          LOC = "R1" ;
NET "mcb3_dram_a[7]"          LOC = "P10" ;
NET "mcb3_dram_a[8]"          LOC = "N4" ;

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NET "mcb3_dram_a[9]"	LOC = "N3" ;
NET "mcb3_dram_ba[0]"	LOC = "P3" ;
NET "mcb3_dram_ba[1]"	LOC = "P1" ;
NET "mcb3_dram_ba[2]"	LOC = "N5" ;
NET "mcb3_dram_cas_n"	LOC = "P8" ;
NET "mcb3_dram_ck"	LOC = "R4" ;
NET "mcb3_dram_ck_n"	LOC = "R3" ;
NET "mcb3_dram_cke"	LOC = "M9" ;
NET "mcb3_dram_dm"	LOC = "W3" ;
NET "mcb3_dram_dq[0]"	LOC = "Y3" ;
NET "mcb3_dram_dq[10]"	LOC = "AE2" ;
NET "mcb3_dram_dq[11]"	LOC = "AE1" ;
NET "mcb3_dram_dq[12]"	LOC = "AD3" ;
NET "mcb3_dram_dq[13]"	LOC = "AD1" ;
NET "mcb3_dram_dq[14]"	LOC = "AB3" ;
NET "mcb3_dram_dq[15]"	LOC = "AB1" ;
NET "mcb3_dram_dq[1]"	LOC = "Y1" ;
NET "mcb3_dram_dq[2]"	LOC = "W2" ;
NET "mcb3_dram_dq[3]"	LOC = "W1" ;
NET "mcb3_dram_dq[4]"	LOC = "T3" ;
NET "mcb3_dram_dq[5]"	LOC = "T1" ;
NET "mcb3_dram_dq[6]"	LOC = "U2" ;
NET "mcb3_dram_dq[7]"	LOC = "U1" ;
NET "mcb3_dram_dq[8]"	LOC = "AA2" ;
NET "mcb3_dram_dq[9]"	LOC = "AA1" ;
NET "mcb3_dram_dqs"	LOC = "V3" ;
NET "mcb3_dram_dqs_n"	LOC = "V1" ;
NET "mcb3_dram_odt"	LOC = "P6" ;
NET "mcb3_dram_ras_n"	LOC = "N8" ;
NET "mcb3_dram_reset_n"	LOC = "L4" ;
NET "mcb3_dram_udm"	LOC = "V4" ;
NET "mcb3_dram_udqs"	LOC = "AC2" ;
NET "mcb3_dram_udqs_n"	LOC = "AC1" ;
NET "mcb3_dram_we_n"	LOC = "P5" ;

#####

#RZQ is required for all MCB designs. Do not move the location #
 #of this pin for ES devices. For production devices, RZQ can be moved to any #
 #valid package pin within the MCB bank. For designs using Calibrated Input Termination, #
 #a 2R resistor should be connected between RZQ and ground, where R is the desired #
 #input termination value. Otherwise, RZQ should be left as a no-connect (NC) pin. #

#####

NET "mcb3_rzq"	LOC = "AC7" ;
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#####

#ZIO is only required for MCB designs using Calibrated Input Termination. #
 #ZIO can be moved to any valid package pin (i.e. bonded IO) within the #
 #MCB bank but must be left as a no-connect (NC) pin. #

#####

NET "mcb3_zio"	LOC = "AE3" ;
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